

YenCare — Scope Sheet

Product 4 · AmaliTech CSR Capstone Internship Programme · Gate 1

Deliverable: a working web or mobile application. Submitting for Gate 1 approval: Friday 4 September 2026.

This sheet sets out our plan for the three build weeks that follow Gate 1 — nothing described here has been built yet.

1. The community user

Patients at the KNUST Clinic in Kumasi who don't own a smartphone, don't have mobile data, or don't have reliable internet. Also the clinic's front-desk staff, who currently handle appointments and queues by hand.

2. The problem, as it happens today

Patients without a smartphone can't book ahead of time. They walk into the clinic, join a physical queue, and wait without knowing how long it will take. Staff track bookings and walk-ins on paper, so it's hard to see who's next or to adjust the queue when a doctor is unavailable.

3. What we are building

YenCare will be an appointment and queue system patients can use through a web app or by SMS. Once it's built, a patient will be able to book ahead, get a text confirming the slot, and see roughly where they stand in the queue instead of just showing up and waiting.

4. Core feature — working end to end by Friday 11 September (Gate 2)

A patient will be able to register and book an appointment through the web app, and get an SMS confirming it. Booking, saving it to the database, and sending the text must all work, start to finish, by Gate 2.

5. Explicitly out of scope

- A USSD version — only considered after Gate 3, if there's time.
- Automatic reminder texts — the MVP sends one confirmation message only.
- Patients rescheduling their own appointment — for now they cancel and rebook manually.
- Automatically rebooking patients when a doctor cancels.
- Staff dashboard showing anything beyond check-in and queue status (no “in progress” or “completed” states).
- An offline mode for the staff dashboard.
- Usage reports or charts, like busiest hours.
- A log of every action staff take.
- More than one backend service or database — we'll use one of each.
- A full test suite for booking clashes — just one test to check double-booking.

6. Who owns what

Name	Track / Role	What they own
Blessing Edmund Kwame Dogbe	DevOps — Team Lead	SMS provider setup (Africa's Talking / Hubtel), hosting, environment config, CI/CD, coordinating the team and weekly check-ins.
Able Kafu Azanda	Backend — Builder	Appointment and queue API, database design, stopping double-bookings, wireframes and the design system.
Emmanuella Lodonu	Backend — Foundations	Patient registration and lookup, and triggering the SMS when someone books.

Harry “Ephraim” Nartey	Fullstack — Foundations	Staff dashboard — check-in and queue view — and connecting the frontend to the backend.
Raymond B. Afrani	Frontend — Foundations	Patient-facing booking form, appointment lookup, and the queue position screen.

7. Offline-first check

Question	Our answer
Does the product work with no or poor internet?	Partly. A patient without internet or a smartphone will still be able to book, get a confirmation, and check the queue — by SMS instead of the web app.
Which parts need a connection, and what happens without one?	The web app will need internet for registering, booking, looking up an appointment, and cancelling. The staff dashboard will also need a connection to show the live queue. We are not planning an offline mode for the MVP.
Any heavy dependency (large downloads, paid API, cloud-only service)?	The SMS provider (Africa's Talking or Hubtel, sandbox account). Everything else is standard and will run on our own machines.
If connectivity fails during the Week 5 demo, how will you still show it?	A short screen recording of the whole flow, plus a phone hotspot as backup internet.

8. Tools and stack

- Backend: Node.js and Express.
- Database: PostgreSQL.
- Frontend: React.
- SMS: Africa's Talking or Hubtel (sandbox account).
- Hosting: Render, Railway, or a basic AWS setup.
- CI: GitHub Actions, to run tests when code is pushed.

9. Milestone plan

By when	What will be true
Fri 11 Sep — Gate 2	A patient can book an appointment on the web app and get an SMS confirmation. Booking, saving it to the database, and sending the text all work start to finish.
Fri 18 Sep — Gate 3	Booking, cancelling, staff check-in, and the live queue position all work. Reminder texts and letting patients reschedule themselves are extras we'll only attempt if there's time left over. Feature freeze after this.
Week 5 — what we demo	Web booking, SMS confirmation, the staff dashboard, and checking the queue — plus our double-booking test and a short setup guide.

10. Risks and assumptions

- Three weeks is tight for five people working part-time, so we're aiming for one thing working well rather than several things half-working.
- The SMS sandbox may have rate limits or delays, and may behave differently from a live service.
- Some of us are starting from template code, which could slow things down early on.
- Two patients could try to book the same slot — we're handling this with database checks and one focused test.
- We need doctor and schedule information from the clinic in time to test with; we'll use placeholder data if it's late.
- At least one of us needs stable internet for hosting and deployment during Gates 1–3.
- The feature freeze after Gate 3 is final — anything not done by then won't be in the demo.

Approval — Programme Lead use only

Decision: Approved / Approved with changes / Resubmit Date: _____

Conditions or changes required:

Signed: _____